

EXAM 7 – SPRING 2011

5. (1.5 points)

Given the following data for a retrospectively rated policy:

Retro Adjustment Period	<u>For Losses Capped at Per Accident Limit</u>		Loss Elimination Ratio from Per Accident Limit
	<u>Insurance Charge at Retro Maximum</u>	<u>Insurance Charge at Retro Minimum</u>	
First	0.113	0.004	4.5%
Second	0.142	0.003	5.5%
Third	0.154	0.003	6.2%
Subsequent	0.158	0.002	6.5%

a. (1 point)

Calculate the cumulative loss capping ratios for each retrospective adjustment period.

b. (0.5 point)

Explain the trend in the loss capping ratios from part a. above.

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Sample Solutions
Exam 07, 2011, Question 5

Solution 1

a)

Retro Adjusted Period (1)	Insurance Charge at Retro Max (2)	Insurance Charge at Retro Min (3)	Eliminated by Retro Max/Min (4) = (2)-(3)	LER from per Accident Limit (5)	Cumulative Loss Capping Ratio (6)
1	.113	.004	.109	.045	.846
2	.142	.003	.139	.055	.806
3	.154	.003	.151	.062	.787
3+	.158	.002	.156	.065	.779

b)

The loss capping ratios given in the problem are decreasing. Over time an increasing proportion of the loss development occurs outside of the loss limitation, lowering the capped losses as a percent of the total loss.